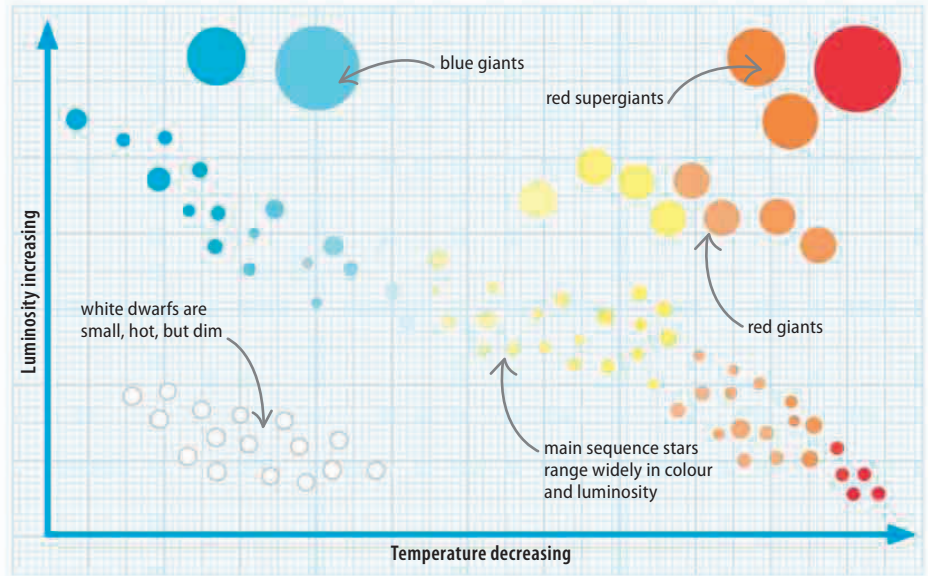


Star types

Stars have a life cycle and, as they age, their characteristics—size, color, luminosity (energy output)—change. Astronomers group stars according to the light they emit. The color of a star's light identifies how hot it is, with blue as the hottest. However, hot stars are not always the brightest. Arranging stars by luminosity and temperature shows they fall into certain groups.

▷ Star groups

When a star first produces energy by nuclear fusion—like our Sun—it is called a main sequence star.



Star life cycle

Stars are born inside great clouds of gas and dust called nebulae. As the clouds collapse, the temperature and pressure rise, and eventually nuclear fusion begins in the star's core. The star then stabilizes on the main sequence of its life. When the fuel runs out, the star dies.

▽ Birth, life, and death

Stars spend around 90 percent of their lives in the main sequence phase. The mass of a star is crucial and will determine what happens to it and when it dies.

